UNIVERSITEIT UTRECHT

MASTER THESIS

Addressing the Cold Start Problem in AI-Assisted Systematic Reviews

Getting started when evidence is scarce

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1 Introduction

1.1 The framework: AI-assisted systematic reviews

Researchers and practitioners are continually challenged to base their decisions on the latest scientific evidence (Sackett et al. 1996; Subbiah 2023; Sheridan and Julian 2016; Chalmers, Hedges, and Cooper 2002). To that end, systematic reviews and meta-analyses were developed as rigorous methods of summarizing scientific literature (Grant and Booth 2009; Brignardello-Petersen, Santesso, and Guyatt 2025; Borenstein et al. 2021; Kicinski 2015). However, systematically reviewing large bodies of literature can be time-consuming. In particular, screening titles and abstracts to identify relevant papers is often a major bottleneck limiting the practical applicability of systematic reviews (Nussbaumer-Streit et al. 2021; Bastian, Glasziou, and Chalmers 2010; Borah et al. 2017).

Fortunately, recent advances in machine learning have produced tools that assist researchers in screening papers by prioritizing the most relevant papers for screening first (Van De Schoot et al. 2021; Ouzzani et al. 2016; Ge et al. 2024; Dos Reis et al. 2023). One such tool is *ASReview*, an open-source software package for AI-assisted systematic reviews (Van De Schoot et al. 2021; J. d. Bruin et al. 2025). ASReview uses active learning models to iteratively learn from the user's screening decisions and adaptively prioritize the most relevant papers for screening. AI-assisted screening can significantly reduce the time and effort needed to conduct systematic reviews and may even help identify more relevant papers compared to traditional screening methods (Teijema et al. 2025; Byrne et al. 2024; Ferdinands et al. 2023; Schoot et al. 2025).

1.2 The problem: the cold start problem

A key challenge limiting the application of AI-assisted systematic reviews is the "cold start" problem (Van De Schoot et al. 2021; Lika, Kolomvatsos, and Hadjiefthymiades 2014). In the context of AI-assisted systematic reviews, the cold start problem refers to the inability of the active learning model to prioritize relevant papers for screening when it has yet to be trained on any examples of relevant papers (Bobadilla et al. 2013; X. Zhang et al. 2023). Notably, however, the cold start problem is usually not an issue because researchers can initialize, or 'warm up', the active learning model using *true examples* of relevant papers prior to screening (Panda and Ray 2022). However, in some cases little to no prior research may be available (Funtowicz and Ravetz 1994; Evans 2022). In this case, the researcher may have to start screening papers at random until a relevant paper has been found which the model can be trained on. The cold start problem is particularly problematic in crisis situations such as COVID-19, where there may not be relevant prior literature and timely evidence synthesis is crucial (Tricco et al. 2020; Kurzer and Ornston 2023).

1.3 The proposed solution: eligibility criteria and large language models

A possible solution to the cold start problem is to use the systematic review's eligibility criteria criteria to initialize the active learning model. The eligibility criteria criteria of a systematic review are typically well-defined and provide a clear description of what constitutes a relevant paper for that review (Cochrane 2026). By using the eligibility criteria criteria as an example of a relevant paper, the active learning model can be trained on this example before screening begins, thus potentially avoiding a cold start. Moreover, it may be possible to further improve starting performance by using large language models (LLMs) to generate synthetic examples of relevant papers based on the eligibility criteria criteria (Riegler et al. 2025). LLMs have been shown to be able to address the cold start problem by leveraging the information that the LLM has been trained on (Bachmann et al. 2025; W. Zhang et al. 2025; Bayer and Reuter 2024). LLMs may therefore be able to generate examples that are more similar to actual abstracts of relevant papers than the eligibility criteria criteria themselves (Steyvers et al. 2025). However, the use LLMs may also add unncessary steps or even negatively impact starting performance by adding noise to the training data of the active learning model (Gartlehner et al. 2025; Sollini et al. 2025).

1.4 Objectives

Our study has two main objectives:

1. To investigate whether the starting performance of AI-assisted systematic reviews can be improved by initialising the AI-model with synthetic abstracts generated by an LLM, based on a systematic review's eligibility criteria criteria or by directly using the eligibility criteria criteria as an example of a relevant paper, compared to a cold start.

2. To investigate whether the use of LLM-generated examples improves starting performance beyond that achieved through initialisation using the systematic review's eligibility criteria criteria directly.

2 Methodology

2.1 Simulation study design

We perform simulations of the abstract-title based screening using ASReview[1] with different types of starting conditions (Van De Schoot et al. 2021; J. d. Bruin et al. 2025; De Bruin et al. 2023). We simulate the abstract-title based screening of 23 previously screened and labelled systematic reviews from the SYNERGY collection.

The four different types of starting conditions are the *LLM* condition, the *cold-start* condition, the *true-example* condition, and the *criteria-only* condition.

In the LLM condition, the eligibility criteria criteria of a published systematic review are provided to OpenAI's GPT-4o-mini model with the request to generate at least one set of titles and abstracts of a paper relevant to the review at hand[2]. These sets of titles and abstracts are then given to ASReview along with the published systematic review to screen. See 1 for a schematic overview of the simulation pipeline.

[1]Two sequential configurations of ASRview are used: the first queries papers at random until at least one relevant and one irrelevant paper have been found and ASReview's classifier can be trained. Next the U4 configuration is used for the remainder of the screening process. Notably, however, in all conditions except for the cold-start condition, the first configuration is effectively skipped since at least one relevant paper is provided before screening begins and irrelevant papers are plentiful in all systematic reviews.

[2]OpenAI's gpt-4o-mini model is instructed via a DSPY module for dynamic prompt generation in each run (Khattab et al. 2023)

In the cold-start condition, no papers are provided prior to the start of screening. In the true-example condition, one relevant paper is sampled from the set of actually relevant papers in the systematic review (with replacement between runs). In the criteria-only condition, the eligibility criteria criteria of the systematic review function directly as an example of an abstract of a relevant paper with which to initialise the LLM, rather than using the LLM to generate an example.

Finally, the eligibility criteria of the systematic reviews are manually manipulated in several ways to investigate whether this affects starting performance. Specifically, the alterations made are: the presence of exclusion criteria, expansion of abbreviations, and expansion of generic terms using MeSH[3].

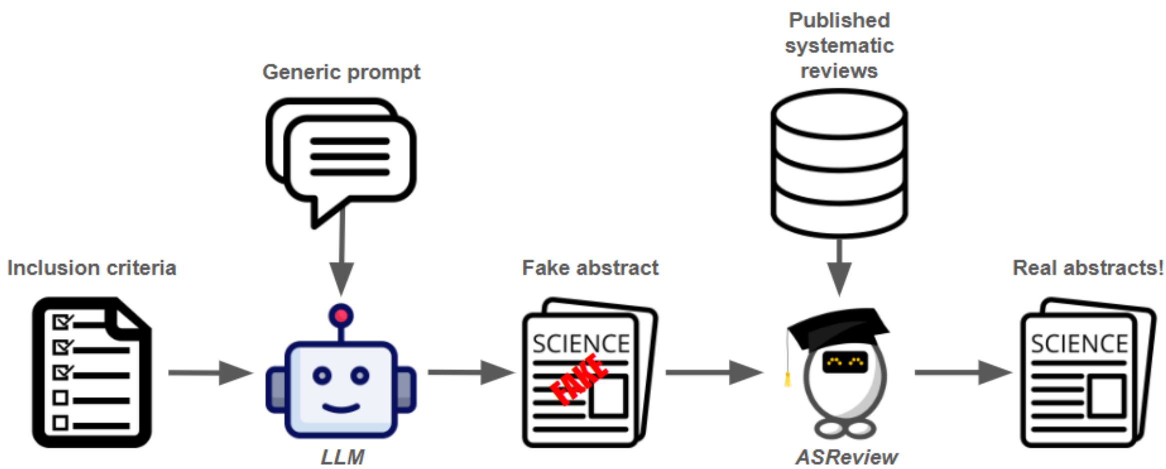


Figure 1: Simulation pipeline (for more detailed schematic of workings within ASReview see (J. d. Bruin et al. 2025))

2.2 Analysis plan

Starting performance will be compared between the four conditions visually using a boxplot and statistically using a linear model with bootstrapped SEs. Starting performance is defined as the number of relevant papers successfully retrieved from among the first 100 papers screened. Some runs may have fewer than 100 screened papers because all relevant papers are retrieved and screening is stopped before the 100th screened paper. Thus, X/n, where $n \leq 100$. The model will include a dummy variable for dataset to account for the fact that some datasets may be easier to screen than others. To specifically address the two objectives of the study described above, three contrasts will be compared for statistical significance differences in starting performance: (1) LLM vs. cold-start, (2) criteria-only vs. cold-start, and (3) LLM vs. criteria-only. An additional analysis will be conducted to explore the impact of manually manipulating the eligibility criteria.

[3]For example, "rodents" may be expanded to include hyponyms such as "mice" and "rats", and "Wilson's disease" to include synonyms such as "Progressive Lenticular Degeneration"

3 Results

3.1 Zoomed in recall plot

Explain what a recall curve is and how it can be interpreted, noting that in the following section only a subpart of the recall curve is shown (i.e., the first 100 screened papers) because we are interested in starting performance.

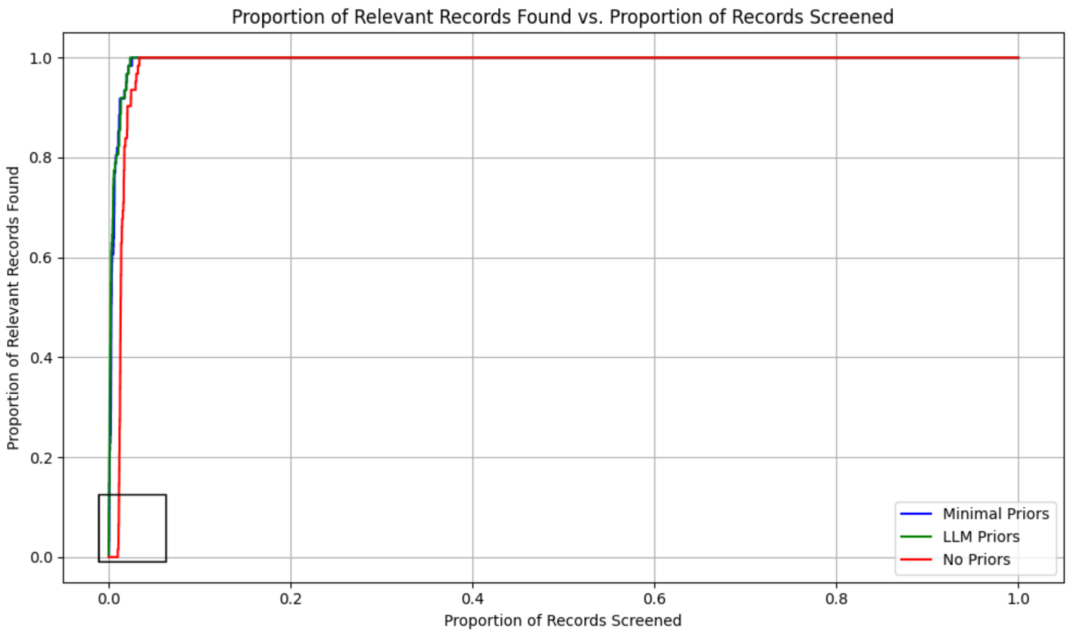
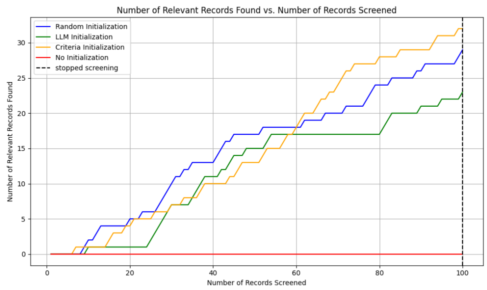


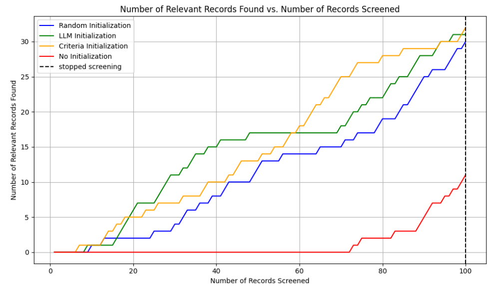
Figure 2: Recall plot of a single run of the true-example condition for the Brouwer (2019) dataset

3.2 Four recall plots of single runs of two datasets per condition (as didactic examples)

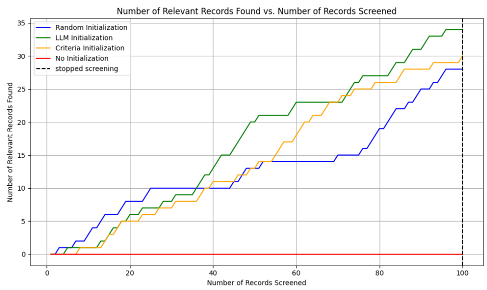
Contain actual data but have been cherry-picked to show the differences between conditions. The goal is to show the reader what the recall plots look like and how they differ between conditions.



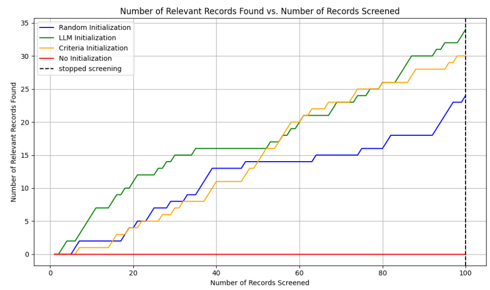
(a) True examples condition (showing two quite differing recall curves)



(b) Cold start condition (showing one entirely flat recall curve and one half flat recall curve)



(c) LLM condition (showing two decent recall curves, similar performance)



(d) Criteria-only condition (showing two decent recall curves, similar performance)

Figure 3: Four recall plots of single runs of two datasets per condition (as didactic examples)

3.3 Aggregate recall plots with line for each condition, for two datasets

The recall plots shown above are just examples of single runs. By simulating many many runs, we can obtain a more reliable estimate of the average performance of each condition. (i.e., do statistical analysis / significance testing).

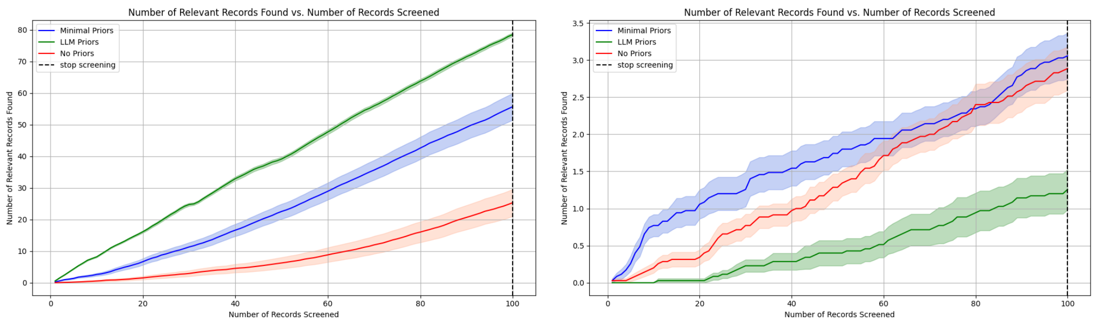


Figure 4: The number of relevant records found per dataset

3.4 Boxplots of starting performance across all runs

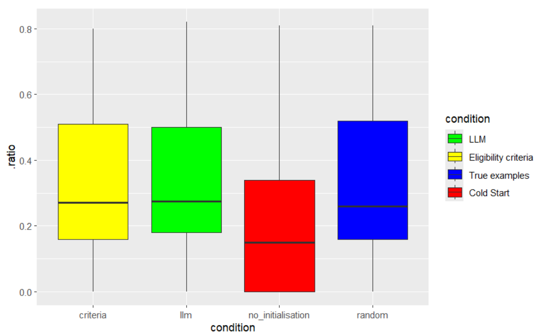


Figure 5: Example boxplot

3.5 Few sentences describing the results of the statistical analysis

One alinea on reporting the statistical significance of the three contrasts: (1) LLM vs. cold-start, (2) criteria-only vs. cold-start, and (3) LLM vs. criteria-only.

3.6 Additional analysis on using eligibility criteria as prior

One alinea reporting the results of manually manipulating / enriching the elgibility criteria.

4 Conclusion

The conclusion is that the cold start problem can be overcome by using the eligibility criteria of a systematic review to initialize the active learning model, and that LLM-generated examples do not improve starting performance beyond that achieved through initialisation using the systematic review's eligibility criteria criteria directly.

5 Discussion

1. A limitating factor in the application of AI-assisted systematic reviews is the cold start problem. This is especially relevant to crisis situations such as COVID-19, where there may not be relevant prior literature and timely evidence synthesis is crucial.

2. Prior research has shown that the cold start problem can be overcome by providing as little as one true pair example of a relevant paper and an irrelevant paper to the active learning model before screening begins (Ferdinands et al. 2023; Teijema et al. 2025). However, in some cases, there may be no prior examples of relevant papers available to initialize the model with. Moreover, the true examples a user has available may not be representative of the relevant papers in the systematic review, which may lead to suboptimal starting performance (i.e., bias / anchor effect – ask Sixu).

3. Currently, the cold start problem is typically addressed by consulting a domain expert to provide true examples of relevant papers to initialize the model with. However, this may not always be feasible, especially in crisis situations where timely evidence synthesis is crucial. (ask Ula – EKE citations).

4. Some limitations of the current study include:

   (a) Only a limited number of historical datasets were used. SYNERGY 2.0 coming soon!

   (b) The of historical datasets. Future studies may consider addressing the cold start problem in a new systematic review.

   (c) ...

5. Practical implications of the current study include:

   (a) If no prior examples of relevant papers are available, the eligibility criteria of a systematic review can be used to initialize the active learning model and overcome the cold start problem.

   (b) Don't use LLMs. LLMs add very little at most and there are many disadvantages to their use.

   (c) If one true example of a relevant paper is available, still use the eligibility criteria to initialize the model. This way your true paper can be used as a test for completeness of your screening (i.e., stopping rule is reached when the true example is found – cite Tale & Josien).

   (d) ...

A SYNERGY metadata

Table 1: Datasets overview

Nr	Dataset	Topic(s)	Records	Included	%
1	Appenzeller-Herzog_2019	Medicine	2873	26	0.9
2	Bos_2018	Medicine	4878	10	0.2
3	Brouwer_2019	Psychology, Medicine	38114	62	0.2
4	Chou_2003	Medicine	1908	15	0.8
5	Donners_2021	Medicine	258	15	5.8
6	Hall_2012	Computer science	8793	104	1.2
7	Leenaars_2019	Psychology, Chemistry, Medicine	5812	17	0.3
8	Leenaars_2020	Medicine	7216	583	8.1
9	Meijboom_2021	Medicine	882	37	4.2
10	Menon_2022	Medicine	975	74	7.6
11	Moran_2021	Biology, Medicine	5214	111	2.1
12	Muthu_2021	Medicine	2719	336	12.4
13	Nelson_2002	Medicine	366	80	21.9
14	Oud_2018	Psychology, Medicine	952	20	2.1
15	Radjenovic_2013	Computer science	5935	48	0.8
16	Sep_2021	Psychology	271	40	14.8
17	Smid_2020	Computer science, Mathematics	2627	27	1.0
18	van_de_Schoot_2018	Psychology, Medicine	4544	38	0.8
19	van_der_Valk_2021	Medicine, Psychology	725	89	12.3
20	van_der_Waal_2022	Medicine	1970	33	1.7
21	van_Dis_2020	Psychology, Medicine	9128	72	0.8
22	Walker_2018	Biology, Medicine	48375	762	1.6
23	Wassenaar_2017	Medicine, Biology, Chemistry	7668	111	1.4
24	Wolters_2018	Medicine	4280	19	0.4

Table 2: Please note that two of the datasets included in the original SYNERGY dataset were excluded entirely due to data quality issues: Chou (2003) and Jeyaraman (2020). For one dataset (Moran, 2021), an updated version was used due to data quality issues in the original version.

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